Patrick Amadeus Irawan

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Website   LinkedIn   Github

Research Interests

My research goal is to build efficient, self-evolving multimodal systems that can acquire new skills and improve after its initial training. I study test-time training, inference-time scaling, continual learning, and self-refinement as mechanisms for learning from data, feedback, and interaction. My prior work spans VLM post-training and multilingual, multicultural, and low-resource multimodal evaluation.

Research Areas: efficient multimodal learning • self-evolving • test-time adaptation • post-training • VLM • multimodal reasoning and evaluation

Education

Ph.D. in Natural Language Processing — MBZUAI *2025 - 2029*
- Advisors: Alham Fikri Aji

Global Study Program — University of California, Davis *2023*
- *computational neuroscience, tech management*

Bachelor in Computer Science — Institut Teknologi Bandung *2020 - 2024*
- *visual question answering; synthetic data generation; reasoning LLM*

Work Experience

Research Engineer *Singapore*
Singapore Management University ***Feb 2025 – Sep 2025***
- Discovered up to a 30% performance degradation in frontier VLMs upon evaluation on a multicultural visual grounding & VQA benchmark, resulting in a benchmark-focused EMNLP publication & analysis-focused preprint.
- Reduced data validation timeline by days for 50K+ spatial annotations with an automated VLM-based evaluation pipeline that automatically flagged poor polygon segmentation via foreground-color ratios.
- Minimized model guessing shortcuts and ensured benchmark robustness by engineering a heuristic-based question-generation engine to synthesize hard adversarial distractors.
- Quantified instance-specific performance decay in Video LLMs by benchmarking temporal moment segmentation in long-horizon culturally-nuanced videos.

Software Engineer *Germany*
IT Bauschmiede ***Nov 2024 – Jul 2025***
- Automated manual enterprise operations by integrating real-time scheduling, invoicing, and geolocation systems serving 3+ scaleup companies.
- Cut system latency by ~20% while maximizing uptime and scalability by architecting a microservices migration from a legacy monolithic MVC infrastructure.

Data Science, SWE, Research Internships
Ruangguru, Blibli, Supertype ***2022-2024***
- Scaled command-line autograder software to support 1k+ concurrent users with 99% uptime.
- Contributed to an 8% increase in absolute approved test-case volume within 6 months at major corporation scale.
- Optimized a topic extraction pipeline, resulting in a 2x increase in speed and a 20% improvement in performance.
- Led the development of real-time AR and RAG multimodal systems for a high-stakes demo of 20k+ audiences.

Publications

*** First author   † Second author**

LinguDistill: Recovering Linguistic Ability in Vision Language Models via Selective Cross-Modal Distillation* — *Preprint* 2026
VLM, SFT, post-training, knowledge distillation, architecture

Anthropogenic Regional Adaptation in Multimodal Vision-Language Model — *Under Review* 2026
VLM, SFT, post-training, data curation, evaluation, multilingual, multimodal

Vision Language Models are Confused Tourists* — CVPR 2026
VLM, visual reasoning, robustness, hallucination, interpretability

A Massive-Scale Multilingual Multi-Cultural Multimodal RAG† — CVPR 2026
RAG, multilingual, multimodal, evaluation, test-time scaling

Counting to Four is Still a Chore for VLMs* — MMFM @ CVPR 2026
VLM, numerical reasoning, interpretability, RL

Entropy2Vec: Crosslingual Language Modeling Entropy as End-to-End Learnable Language Representations* — MRL @ EMNLP 2025
LLM, multilingual, representation learning, unsupervised learning

WorldCuisines: A Massive-Scale Benchmark for Multilingual and Multicultural VQA on Global Cuisines* — NAACL 2025 (Best Theme Paper 🏆)
VQA, multilingual, multicultural, benchmark, evaluation, data curation

Seeing Culture: A Benchmark for Visual Reasoning and Grounding† — EMNLP 2025
visual grounding, evaluation, object detection, multicultural

Predicting Language Model Performance on Multilingual Tasks via Proxy Models — NAACL 2025
multilingual, evaluation, efficient, regression, SFT

Towards Efficient and Robust VQA-NLE Data Generation with Large Vision-Language Models* — COLING 2025
VQA, VLM, NLP, synthetic data generation, visual reasoning

SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages — EMNLP 2024
multilingual, multicultural, video understanding, speech understanding, evaluation

Achievement & Awards

OpenAI Micro Grant Award **Grantee**	*2025*
NAACL Best Theme Paper Award **Winner**	*2025*
BCG SEA Emeralds Case Competition **Winner**	*2024*
Indonesian International Student Mobility Awards @ UC Davis **Awardee**	*2023*
DOMO Higher Education Case Competition **Finalist**	*2023*
CSLeaders Full Bachelor Tuition Scholarship **Awardee**	*2022*
Gemastik XV Data Mining Division **Finalist**	*2022*

Advising & Mentorship

- **Rava Maulana** — world models and evaluation (*now Software Engineer at Tiket*)
- **Iskandar Muda** — world models and evaluation (*now M.Sc. student at Bandung Institute of Technology*)
- **Qinrong Cui** — test-time adaptation for fine-grained visual understanding (*now M.Sc. student at MBZUAI*)
- **Wilfried Ariel** — multicultural object detection and evaluation (*now Ph.D. student at NTU*)

References

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